Kubernetes Deployment and Rollout concept

1) CREATE NAMESPACE YAML FILE:

vi rithika-ns.yaml

==> This YAML file typically contains the namespace configuration for Kubernetes.

```
root@ip-172-31-30-2:/home/ubuntu/rithika-k8s# cat rithika-ns.yaml
apiVersion: v1
kind: Namespace
metadata:
  name: rithika-ns
```

2)CREATE DEPLOYMENT YAML FILE:

vi rithika-deployment.yaml

==> `deployment.yaml` contains the Kubernetes Deployment configuration, including replicas, Pod template, container images, ports, and labels.

```
root@ip-172-31-30-2:/home/ubuntu/rithika-k8s# cat rithika-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: wipro-deployment
  namespace: rithika-ns
spec:
  replicas: 3
  selector:
    matchLabels:
      app: wipro
  template:
    metadata:
      labels:
        app: wipro
    spec:
      containers:
      - name: wipro-container
        image: nginx:latest
        ports:
        - containerPort: 80
```

3) **kubectl apply -f rithika-deployment.yaml**

==> Applies the configuration defined in the `deployment.yaml` file to the Kubernetes cluster.

==> It update the resources without manually changing them.

==> It ensures the cluster state matches the YAML specification.

```
root@ip-172-31-30-2:/home/ubuntu/rithika-k8s# kubectl apply -f rithika-deployment.yaml
deployment.apps/wipro-deployment created
```

4) **kubectl get all -n rithika-ns**

==> Lists all Kubernetes resources in the specified namespace ie., rithika-ns.

==> It will quickly see the status of all objects in a namespace.

```
root@ip-172-31-30-2:/home/ubuntu/rithika-k8s# kubectl get all -n rithika-ns
NAME                                   READY   STATUS    RESTARTS   AGE
pod/wipro-deployment-f5d9dcc9b-76w62   1/1     Running   0          66s
pod/wipro-deployment-f5d9dcc9b-nbwv8   1/1     Running   0          66s
pod/wipro-deployment-f5d9dcc9b-pskhs   1/1     Running   0          66s

NAME                               READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/wipro-deployment   3/3     3            3           66s

NAME                                         DESIRED   CURRENT   READY   AGE
replicaset.apps/wipro-deployment-f5d9dcc9b   3         3         3       66s
```

==> Here we can see unique name for each Pod automatically generated by the Deployment.

==> `Running` indicates the Pod is active and ready to serve requests.

==> There are 3 Pods running for this Deployment, meaning the desired number of replicas is met.

5) **kubectl describe deployment wipro-deployment -n rithika-ns**

==> This command shows the detailed information about the Deployment

```
root@ip-172-31-30-2:/home/ubuntu/rithika-k8s# kubectl describe deployment wipro-deployment -n rithika-ns
Name:                   wipro-deployment
Namespace:              rithika-ns
CreationTimestamp:      Thu, 19 Feb 2026 14:07:29 +0000
Labels:                 <none>
Annotations:            deployment.kubernetes.io/revision: 1
Selector:               app=wipro
Replicas:               3 desired | 3 updated | 3 total | 3 available | 0 unavailable
StrategyType:           RollingUpdate
MinReadySeconds:        0
RollingUpdateStrategy:  25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=wipro
  Containers:
   wipro-container:
    Image:         nginx:latest
    Port:          80/TCP
    Host Port:     0/TCP
    Environment:   <none>
    Mounts:        <none>
  Volumes:         <none>
  Node-Selectors:  <none>
  Tolerations:     <none>
Conditions:
  Type           Status  Reason
  ----           ------  ------
  Available      True    MinimumReplicasAvailable
  Progressing    True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   wipro-deployment-f5d9dcc9b (3/3 replicas created)
Events:
  Type    Reason             Age    From                   Message
  ----    ------             ----   ----                   -------
  Normal  ScalingReplicaSet  3m54s  deployment-controller  Scaled up replica set wipro-deployment-f5d9dcc9b from 0 to 3
```

6) **Here i am changing the versions in the yaml file :**

==> Initially deployed an application using nginx : latest version, now i am updated with a new nginx:1.25 version. Kubernetes will creates a new ReplicaSet for the updated Pods.

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: wipro-deployment
  namespace: rithika-ns
spec:
  replicas: 3
  selector:
    matchLabels:
      app: wipro
  template:
    metadata:
      labels:
        app: wipro
    spec:
      containers:
      - name: wipro-container
        image: nginx:latest
        ports:
        - containerPort: 80
```

```yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: wipro-deployment
  namespace: rithika-ns
spec:
  replicas: 3
  selector:
    matchLabels:
      app: wipro
  template:
    metadata:
      labels:
        app: wipro
    spec:
      containers:
      - name: wipro-container
        image: nginx:1.25
        ports:
        - containerPort: 80
```

7) **kubectl describe pod wipro-deployment- 5dd5786f6c -n rithika-ns**

==> This command shows the detailed information about a specific Pod.

==> Here i am getting the details of new updated version.

```
root@ip-172-31-30-2:/home/ubuntu/rithika-k8s# kubectl describe pod wipro-deployment-5dd5786f6c   -n rithika-ns
Name:             wipro-deployment-5dd5786f6c-5h4p8
Namespace:        rithika-ns
Priority:         0
Service Account:  default
Node:             ip-192-168-31-61.ap-south-1.compute.internal/192.168.31.61
Start Time:       Thu, 19 Feb 2026 14:22:43 +0000
Labels:           app=wipro
                  pod-template-hash=5dd5786f6c
Annotations:      <none>
Status:           Running
IP:               192.168.31.206
IPs:
  IP:           192.168.31.206
Controlled By:  ReplicaSet/wipro-deployment-5dd5786f6c
Containers:
  wipro-container:
    Container ID:   containerd://6e5ccd969542dbfbb9d3c7c9d6b6e3e04ecfe7d117b5c496401ed8b5e9e67eb2
    Image:          nginx:1.25
    Image ID:       docker.io/library/nginx@sha256:a484819eb60211f5299034ac80f6a681b06f89e65866ce91f356ed7c72af059c
    Port:           80/TCP
    Host Port:      0/TCP
    State:          Running
      Started:      Thu, 19 Feb 2026 14:22:49 +0000
    Ready:          True
    Restart Count:  0
    Environment:    <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-kqjkk (ro)
Conditions:
  Type                        Status
  PodReadyToStartContainers   True
  Initialized                 True
  Ready                       True
  ContainersReady             True
  PodScheduled                True
```

8) **kubectl get all -n rithika-ns**

==> Here it the previous version

```
root@ip-172-31-30-2:/home/ubuntu/rithika-k8s# kubectl get all -n rithika-ns
NAME                                  READY   STATUS    RESTARTS   AGE
pod/wipro-deployment-f5d9dcc9b-76w62   1/1     Running   0          14m
pod/wipro-deployment-f5d9dcc9b-nbwv8   1/1     Running   0          14m
pod/wipro-deployment-f5d9dcc9b-pskhs   1/1     Running   0          14m

NAME                               READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/wipro-deployment   3/3     3            3           14m
```

==> Kubernetes keeps old ReplicaSets for rollback purposes, but only the new ReplicaSet is currently active.

==> After updating the version here new ReplicaSet created after updating the Deployment:

```
root@ip-172-31-30-2:/home/ubuntu/rithika-k8s# kubectl get all -n rithika-ns
NAME                                    READY   STATUS    RESTARTS   AGE
pod/wipro-deployment-5dd5786f6c-5h4p8   1/1     Running   0          3m49s
pod/wipro-deployment-5dd5786f6c-g4x2w   1/1     Running   0          4m5s
pod/wipro-deployment-5dd5786f6c-zqnpz   1/1     Running   0          3m57s

NAME                               READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/wipro-deployment   3/3     3            3           19m

NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/wipro-deployment-5dd5786f6c   3         3         3       4m5s
replicaset.apps/wipro-deployment-f5d9dcc9b    0         0         0       19m
```

9) **kubectl rollout undo deployment wipro-deployment -n rithika-ns**

==> It is used to rolls back the Deployment to the previous revision.

```
root@ip-172-31-30-2:/home/ubuntu/rithika-k8s# kubectl rollout undo deployment wipro-deployment -n rithika-ns
deployment.apps/wipro-deployment rolled back
```

==> Here it is rolled back.

10) **kubectl get all -n rithika-ns**

==> We can see here it is the new version change we made , after rolled back it will moves to the old version.

```
NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/wipro-deployment-5dd5786f6c   3         3         3       4m5s
replicaset.apps/wipro-deployment-f5d9dcc9b    0         0         0       19m
```

==> here it is rolled back to the older version

```
root@ip-172-31-30-2:/home/ubuntu/rithika-k8s# kubectl get all -n rithika-ns
NAME                                   READY   STATUS    RESTARTS   AGE
pod/wipro-deployment-f5d9dcc9b-96ck9   1/1     Running   0          7m15s
pod/wipro-deployment-f5d9dcc9b-fj62p   1/1     Running   0          7m10s
pod/wipro-deployment-f5d9dcc9b-tb5gg   1/1     Running   0          7m12s

NAME                               READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/wipro-deployment   3/3     3            3           33m

NAME                                          DESIRED   CURRENT   READY   AGE
replicaset.apps/wipro-deployment-5dd5786f6c   0         0         0       18m
replicaset.apps/wipro-deployment-f5d9dcc9b    3         3         3       33m
```

11) **kubectl rollout history deployment wipro-deployment -n rithika-ns**

==> By this command it shows all revisions of a Deployment.

```
root@ip-172-31-30-2:/home/ubuntu/rithika-k8s# kubectl rollout history deployment wipro-deployment -n rithika-ns
deployment.apps/wipro-deployment
```

==> we can see the old and updated version here

```
REVISION  CHANGE-CAUSE
1         <none>  # old version
2         <none>  # updated version
```

12) **kubectl rollout status deployment wipro-deployment -n rithika-ns**

==> This command shows the current rollout status of a Deployment.

==> It verifies if all Pods are updated and ready after an update or rollback.

```
root@ip-172-31-30-2:/home/ubuntu/rithika-k8s#  kubectl rollout status deployment wipro-deployment -n rithika-ns
deployment "wipro-deployment" successfully rolled out
```